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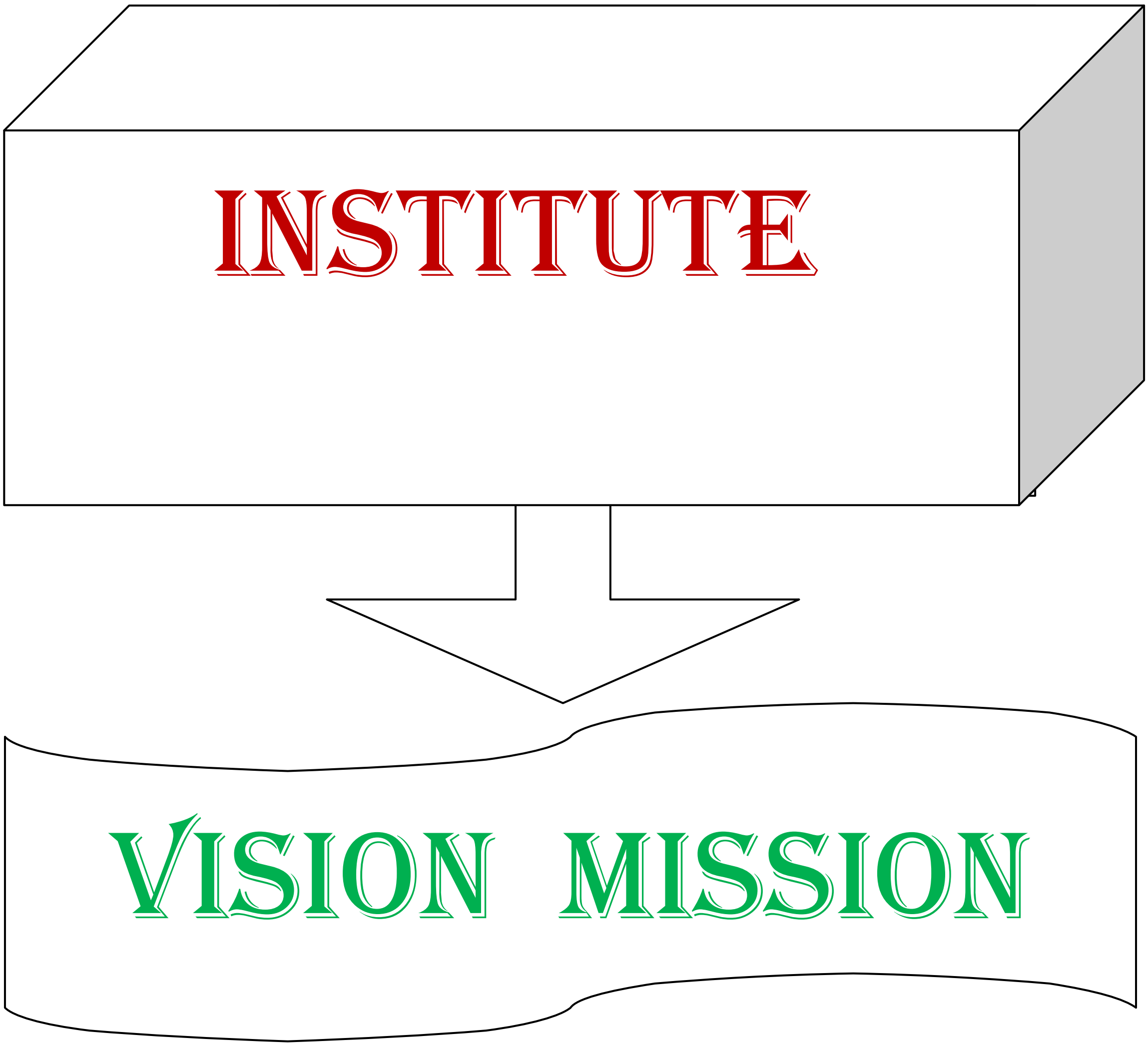
INSTITUTE
VISION MISSION

INSTITUTE
VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning..

DEPARTMENT

VISION MISSION

DEPARTMENT

VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

PROGRAM EDUCATIONAL OBJECTIVES, PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations**.** **[K3]**
6. **The Engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication**: Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance**: Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

Website: www.jntuk.edu.in
Email: dap@jntuk.edu.in

Phone: 0884-2300991

Directorate of Academic Planning
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA
KAKINADA-533003, Andhra Pradesh, INDIA
(Established by AP Government Act No. 30 of 2008)

Lr. No. DAP/RAC/ II,III & IV Year /B. Tech/B. Pharmacy/2021 *Date 08.10.2021*

Dr. R. Srinivasa Rao,
Director, Academic Planning
JNTUK, Kakinada

To
All the Principals of Affiliated Colleges,
JNTUK, Kakinada.

Revised Academic Calendar for II, III, IV Year - B. Tech/B. Pharmacy for the AY 2021-22
(As per G.O. Rt. No. 242, Higher Education (U.E) Dept., dated 13.09.2021)

I SEMESTER			
Description	**From**	**To**	**Weeks**
Commencement of Class Work	**01.10.2021**		
I Unit of Instruction	01.10.2021	20.11.2021	7W
I Mid Examinations	22.11.2021	27.11.2021	1W
II Unit of Instructions	29.11.2021	15.01.2022	7W
II Mid Examinations	17.01.2022	22.01.2022	1W
Preparation & Practicals	24.01.2022	29.01.2022	1W
End Examinations	31.01.2022	12.02.2022	2W
Commencement of II Semester Class Work	**14.02.2022**		
II SEMESTER			
I Unit of Instructions	**14.02.2022**	02.04.2022	7W
I Mid Examinations	04.04.2022	09.04.2022	1W
II Unit of Instructions	11.04.2022	28.05.2022	7W
II Mid Examinations	30.05.2022	04.06.2022	1W
Preparation & Practicals	06.06.2022	11.06.2022	1W
End Examinations	13.06.2022	25.06.2022	2W
Commencement of next Year Class Work			
Note: Calendar is prepared with 8 hrs/day hence 7 weeks per instruction period			

R. Srinivasa Rao
Director Academic Planning
Director
Academic Planning
JNTUK Kakinada

Copy to the Secretary to the Hon'ble Vice Chancellor, JNTUK
Copy to Rector, Registrar, JNTUK
Copy to Director Academic Audit, JNTUK
Copy to Director of Evaluation, JNTUK

CLASS CONSOLIDATED TIME TABLE

Class: VIII Semester W.E.F: 14-02-2022

Sec : A Class Coordinator : D S L Manikanteswari Room No :G-301

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20AM-12.10PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM-04.30 PM)
Mon	SPM	SPM	CS	CS	LUNCH BREAK	BEPG	BEPG	Library
Tue	CS	CS	BEPG	BEPG		SPM	SPM	Sports
Wed	Project Work (Part-B)					Project Work (Part-B)		

Sec : B Class Coordinator : Mr. M Babu Rao Room No : G-301

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20AM-12.10PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM-04.30 PM)
Thus	CS	CS	SPM	SPM	LUNCH BREAK	BEPG	BEPG	Library
Fri	SPM	SPM	BEPG	BEPG		CS	CS	Sports
Sat	Project Work (Part-B)					Project Work (Part-B)		

Sec : C Class Coordinator : A. Rajesh Room No :G-302

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20AM-12.10PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM-04.30 PM)
Mon	SPM	SPM	CS	CS	LUNCH BREAK	BEPG	BEPG	Library
Tue	CS	CS	BEPG	BEPG		SPM	SPM	Sports
Wed	Project Work (Part-B)					Project Work (Part-B)		

Sec : D Class Coordinator : B.Kiran Kumar Room No :G-302

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20AM-12.10PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM-04.30 PM)
Thus	CS	CS	SPM	SPM	LUNCH BREAK	BEPG	BEPG	Library
Fri	SPM	SPM	BEPG	BEPG		CS	CS	Sports
Sat	Project Work (Part-B)					Project Work (Part-B)		

STAFF DETAILS

S. No.	Course Code	Course Name	Sections			
			A	B	C	D
1.	V18CST36	Software Project Management (SPM) - ***(Elective-V)***	D S L Manikanteswari		A.Rajesh	
2.	V18CST43	Cyber Security (CS) - ***(Elective-VI)***	Mr. M Babu Rao		Mr. B Kiran Kumar	
3.	V18EEOE8	Basics of Electrical Power Generation (BEPG) - ***(Open Elective –III)***	Mr. K Venkata Reddy		Mr.Durga R Nookesh	
4.	V18CSP02	Project Work (Part-B)	Mr. M. Satyanarayana Reddy	Mrs. N. Hiranmayee	Mr. M S Kumar Reddy	Dr. P. LaxmiKanth

Head of the Department

Head of the Department
Dept. of Computer Science & Engineering
Sri Vasavi Engineering College
TADEPALLIGUDEM-534 101

COURSE STRUCTURE

VIII - SEMESTER

S.No.	Course Code	Course	L	T	P	C
1	**Elective – V**					
	V18CST36	1. Software Project Management	3	0	0	3
	V18CST37	2. Big Data Analytics				
	V18CST38	3. Soft Computing				
	V18CST39	4. Cloud Computing				
2	**Elective – VI**		3	0	0	3
	V18CST40	1. Software Architecture and Design Patterns				
	V18CST41	2. Middleware Technologies				
	V18CST42	3. Natural Language Processing				
	V18CST43	4. Cyber Security				
3	**Open Elective – III** **(Interdisciplinary)**	**OPE III(1-3)**	3	0	0	3
4	V18CSP02	Project Work (Part-B)	0	0	16	8
Total			**9**	**0**	**16**	**17**

Total Contact Hours: 25

LESSON PLANS

Software Project Management

Academic Year : 2021-22 **Programme**: B.Tech
Semester : VIII **Sections** :A,B,C&D
Name of the Course: **Software Project Management** **Course Code**: **V18CST36**

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO1	Define Software Project Management Terminology and Methodology.	[K1]
2.	CO2	Describe various Software Lifecycle Models, Process Artifacts and Workflows.	[K2]
3.	CO3	Explain various Effort Estimation Techniques for Project Planning.	[K2]
4.	CO4	Demonstrate Risk Management Concepts.	[K3]
5.	CO5	Develop Project Status Reports for tracking and controlling Software Deliverables.	[K3]
6.	CO6	Describe Software Quality Metrics.	[K2]

Text Books:

1. Software Project Management, Bob Hughes & Mike Cotterell, TMH
2. Software Project Management, Walker Royce, Pearson Education, 2005.
3. Software Project Management in Practice, Pankaj Jalote, Pearson

Reference Books:

1. Software Project Management, Joel Henry, Pearson Education.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	75	70	70	65	70	65
	Level 2	70	65	65	60	65	60
	Level 1	65	60	60	55	60	55

Lecture Plan:

Unit-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Dissemination of Vision, Mission of the Dept and PEOs, Pos,& PSOs of the Programme			Lecture	BB
2	**Define Software Project Management Terminology and Methodology (K1)**	Define Software Project Management Terminology. And comparison with other projects	K1	1	Lecture	BB
3		Describe software project management activities.	K1	1	Lecture	BB
4		Describe various Categories in software Projects	K1	1	Lecture	BB
5		Identify types of stake holders, objectives and goals in software project management.	K1	2	Lecture	BB
6		Describe Stepwise project planning and project scope and Objectives.	K1	1	Lecture	BB
7		Identify Project products and Deliverables.	K1	1	Lecture	BB
8		Outline Effort Estimation and Infrastructure.	K1	1	Lecture	BB+ICT

Unit- 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**Describe various Software Lifecycle Models, Process Artifacts and Workflows (K2)**	Outline various Life Cycle Models.	K1	1	Lecture	BB
2		Classify technologies: Process Models	K2	2	Lecture	BB
3		Describe Software Prototyping.	K2	1	Lecture	BB
4		Explain Iterative and Incremental Process Framework.	K2	1	Lecture	BB
5		Classify Project Life Cycle Phases.	K2	2	Lecture	BB+ICT
6		Explain various Artifacts of Software Process.	K2	2	Lecture	BB
7		Explain Process Workflows.	K2	2	Lecture	BB

Unit-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**Explain various Effort Estimation Techniques for Project Planning (K2)**	Describe Software Effort Estimation Techniques.	K1	1	lecture	BB
2		Discuss Function Point Analysis.	K2	1	lecture with Discussion	BB
3		Explain SLOC: Software Metrics and Measurements.	K2	2	lecture	BB + ICT
4		Describe COCOMO: A Parametric Model	K2	2	lecture	BB + ICT
5		Discuss Use-Case based Estimation Techniques.	K2	1	lecture with Discussion	BB
6		Explain various Activity Identification Approaches: Sequencing and Scheduling Activities.	K2	2	lecture	BB
7		Discuss Network Planning Models in Project Scheduling: Critical Path Analysis.	K2	2	lecture with Discussion	BB

Unit- 4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**Demonstrate Risk Management Concepts (K3)**	Describe various Risk Management Categories.	K1	2	Lecture	BB
2		Discuss concepts of Risk Identification, Assessment, Planning and Management.	K2	2	Lecture with discussion	BB+ICT
3		Demonstrate PERT Technique.	K3	1	Lecture	BB
4		Explain Monte Carlo Method for project estimation.	K2	1	Lecture	BB
5		Describe Resource Allocation types	K2	1	Lecture	BB

Unit-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**Develop Project Status Reports for tracking and controlling Software Deliverables (K3)**	Describe the concept of Project Monitoring and Control.	K1	1	lecture	BB
2		Explain Progress Monitoring, and Cost Monitoring in Project Control.	K2	2	lecture	BB
3		Explain Earned Value Analysis in Cost Monitoring.	K2	2	lecture	BB + ICT
4		Discuss various Defects and Issues in Project Monitoring and Control.	K2	1	Lecture	BB + ICT
5		Develop Project Status Reports with Sample Case Study.	K3	1	lecture	BB
6		Discuss various types of resources and resource requirements in Software Project Management.	K2	2	lecture with discussion	BB
7		Explain the concept of Resource Allocation and Scheduling.	K2	1	Lecture with practical	BB

Unit-6

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**Describe Software Quality Metrics(K2)**	Define Software Quality and Quality Factors.	K1	1	Lecture	BB + ICT
2		Explain Software Quality Planning.	K2	1	Lecture	BB + ICT
3		Outline various Quality Measures in Software Quality Management.	K1	2	Lecture	BB + ICT
4		Discuss Quantitative Approaches to Quality Management.	K2	2	Lecture with Discussion	BB + ICT
5		Describe importance of quality and ISO 9126.	K2	1	Lecture	BB + ICT
6		Explain the concepts of product Quality and Process Quality.	K2	1	Lecture	BB + ICT
7		Describe Statistical Process Control Capability Maturity Model.	K2	2	Lecture with Discussion	BB + ICT
8		Discuss various Techniques to Enhance Software Quality.	K2	2	Lecture with Discussion	BB + ICT

Total No. of Classes: 60

Cyber Security

Academic Year : 2021-22
Programme: B.Tech
Year/ Semester: VIII Sem
Section:A, B, C, D
Name of the Course: Cyber Security
Course Code: V18CST43/C409E4

Course Outcomes (Along with knowledge level): At the end of the course, student will be able to:

S. No.	CO No.	Course Outcome	BTL
1.	CO1	Describe about Cybercrimes.	K2
2.	CO2	Explain Cyber criminals and their attacks.	K2
3.	CO3	Illustrate Cybercrimes and security in mobile devices	K2
4.	CO4	Discuss about the Tools and methods used to overcome Cybercrimes.	K2
5.	CO5	Discuss about Cyber Laws and IT Acts.	K2
6.	CO6	Explain about Computer Forensics.	K2

Text Books:

1. Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, NinaGodbole, SunitBelapure, 1stedition, Wiley.

Reference Books:

1. Principles of Information Security, MichealE.Whitman and Herbert J.Mattord, 4th edition, Cengage Learning.
2. Information Security the complete reference, Mark Rhodes, Ousley, 2nd edition, MGH.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		65	65	65	60	60	60
Targeted level of Attainment	Level 3	65	65	65	60	60	60
	Level 2	55	55	55	50	50	50
	Level 1	45	45	45	40	40	40

Lecture Plan:

Unit-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	Describe about Cybercrimes (K2)	Discuss Blooms Taxonomy	K1 and K2	1	Lecture	BB+ICT
2		Explain Cyber Crime	K2	1	Lecture	BB
3		Define about Cyber Crime Describe origins of the Word	K1 and K2	1	Lecture	BB
4		Define Information Security and also Discuss about Cybercrime and Information Security	K1 and K2	1	Lecture	BB
5		Express who are Cybercriminals?	K2	1	Lecture	BB
6		Classifications of Cybercrimes	K2	1	Lecture With Discussion	BB

7		Explain about Cybercrime: The Legal Perspectives	K2	1	Lecture With Discussion	BB
8		Explain about Cybercrimes: An Indian Perspective. Cybercrime and the Indian ITA 2000	K2	1	Lecture	BB
9		Describe the Global Perspective on Cybercrimes.	K2	1	Lecture	BB
10		Discuss about Cybercrime Era: Survival Mantra for the Netizens.	K2	1	Lecture	BB
		TOTAL	**10**			

Unit-2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	Explain Cyber criminals and their attacks (K2)	List about the planning of criminals	K1	1	Lecture	BB
2		Identify how criminals Plan the Attacks	K2	2	Discussion	ICT+BB
3		Explain about Social Engineering	K2	1	Discussion	ICT+BB
4		Discuss about Cyber stalking	K2	1	Discussion	ICT+BB
5		Explain about Cyber cafe and Cybercrimes	K2	2	Discussion	ICT+BB
6		Discuss about Botnets: The Fuel for Cybercrime	K2	1	Discussion with Lecture	BB
7		Attack Vector Cloud Computing	K2	2	Discussion with Lecture	ICT+BB
		TOTAL	**10**			

Unit-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	Illustrate Cybercrimes and security in mobile devices (K2)	Define about wireless devices	K1	1	Lecture	BB
2		Explain the Proliferation of Mobile and Wireless Devices	K2	1	Lecture	BB
3		Discuss the trends in Mobility	K2	1	Lecture	BB+ICT
4		Demonstrate Credit Card Frauds in Mobile and Wireless Computing Era	K2	1	Lecture	ICT
5		Explain the Security Challenges Posed by Mobile Devices	K2	1	Lecture	BB
6		Describe the Registry Settings	K2	1	Discussion	BB+ICT

		for Mobile Devices				
7		Define Authentication and Explain the Authentication Service Security	K1 and K2	1	Lecture	BB
8		Discuss Attacks on Mobile/Cell Phones	K2	1	Lecture	ICT
9		Explain about Mobile Devices Security Implications for Organizations	K2	1	Lecture	BB
10		Discuss about Organizational Measures for Handling Mobile	K2	1	Lecture	ICT
11		Express the Organizational Security Policies and Measures in Mobile Computing Era, Laptops.	K2	1	Lecture with Discussion	BB
		TOTAL	**11**			

Unit-4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	Discuss about the Tools and methods used to overcome Cybercrimes (K2)	Describe about Proxy Servers and Anonymizers	K2	1	Lecture	BB
2		Explain about Phishing, Password Cracking	K2	1	Lecture With Discussion	BB
3		Name the Key loggers and Spywares	K1	1	Lecture With Discussion	BB+ICT
4		Describe about Virus and Worms, Trojan Horses and Backdoors	K2	1	Lecture	BB
5		Discuss about Steganography , DoS and DDoS Attacks	K2	2	Discussion	BB
6		Explain about SQL Injection , Buffer Overflow	K2	1	Discussion	BB
7		Identify the Attacks on Wireless Networks	K2	1	Lecture	BB
8		Phishing and Identity Theft	K2	1	Lecture	BB
		TOTAL	**9**			

Unit-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	Discuss about Cyber Laws and IT Acts **(K2)**	Describe about the Indian Context of cyber laws, the Indian IT Act	K2	1	Lecture	BB
2		Summarize the Challenges to Indian Law and Cyber crime Scenario in India	K2	1	Lecture	BB
3		Discuss about the Consequences of Not Addressing the Weakness in Information Technology Act	K2	1	Lecture	BB
4		Discuss about the Digital Signatures and the Indian IT Act	K2	1	Lecture	BB
5		Explain about Information Security Planning and Governance.	K2	1	Discussion	BB+ICT
6		Discuss the Information Security Policy Standards, Practices.	K2	1	Lecture	BB
7		Describe the information Security Blueprint , Security education.	K2	1	Lecture	BB
8		Explain about training and awareness program, Continuing Strategies.	K2	1	Lecture	BB+ICT
		TOTAL	**8**			

Unit-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	Explain about Computer Forensics (K2)	Define Computer forensics	K1	1	Lecture	BB+ICT
2		Discuss the Historical Background of Cyber forensics, Digital Forensics Science	K2	1	Lecture	BB
3		Describe about The Need for Computer Forensics, Cyber forensics and Digital Evidence	K2	1	Lecture	BB
4		Discuss about forensics Analysis of E-Mail.	K2	1	Lecture And Discussion	BB+ICT
5		Explain about Digital Forensics Life Cycle , Chain of Custody Concept	K2	1	Lecture	BB+ICT
6		Describe about network	K2	1	Lecture	BB

		Forensics, Approaching a Computer Forensics Investigation				
7		Discuss about computer Forensics and Steganography	K2	1	Lecture With Discussion	BB
8		Explain about relevance of the OSI 7Layer Model to Computer Forensics	K2	1	Lecture	BB+ICT
9		Describe about the Forensics and Social Networking Sites :The Security/Privacy Threats	K2	1	Lecture	BB
10		Explain about computer Forensics from Compliance Perspective	K2	1	Lecture With Discussion	BB
11		Describe about the challenges in Computer Forensics, Special Tools and Techniques	K2	1	Lecture	BB+ICT
12		Discuss about forensics Auditing, Ant forensics	K2	1	Lecture	BB
		Total	**12**			

Total Number of Hour's Required = 60

Basics of Electrical Power Generation

Academic Year: 2021-22 **Programme:** B.Tech

Year/ Semester: VIII **Section:** A,B,C,D

Name of the Course: Basics of Electrical Power Generation

Course Code : **V18EETOE8**

Course Outcomes (Along with Knowledge Level):

After successful completion of course the student will able to

CO No.	Course Outcome	BTL
CO1	Understand the various energy sources, substations and switchgear devices	(K2)
CO2	Understand the principle of operation of different components of thermal power stations	(K2)
CO3	Understand the principle of operation of different components of nuclear power stations	(K2)
CO4	Understand the principle of operation of different components of hydro power stations	(K2)
CO5	Understand the working of solar photo voltaic systems and applications	(K2)
CO6	Understand the wind energy conversion systems, efficiency and power generation	(K2)

Text Books:

1. A Text Book on Power System Engineering by M. L. Soni, P. V. Gupta, U. S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co. Pvt. Ltd.- 2nd edition, 2013.
2. Renewable Energy Resources, John Twidell and TonyWeir, Taylorand Francis-2nd edition, 2013.

Reference Books:

1. Elements of Electrical Power Station Design by – M V Deshpande, PHI, NewDelhi-3rdedition,2010.
2. Renewable Energy – Edited by Godfrey Boyle – oxford university Press, 3rdedition, 2013.
3. Electrical Power Systems by C. L. Wadhwa, 6th Edition, New Age International Publishers, 2018.
4. Non-conventional energy source–B.H.khan -TMH-2nd edition,2017.
5. https://nptel.ac.in/content/storage2/courses/108105053/pdf/L-02(TB)(ET)%20((EE)NPTEL).pdf

	CO1	CO2	CO3	CO4	CO5	CO6
Proficiency Level	60%	60%	60%	60%	60%	60%
Attainment Level	60%	60%	60%	60%	60%	60%

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
UNIT-I: FUNDAMENTALS OF ELECTRICAL POWER SYSTEM						
1	Understand the various energy sources, substations and switchgear devices (K2)	Explain present Energy scenario (world and India)	K2	2	Lecture & Discussion	Blackboard & PPT
		Explain various Conventional and non-conventional energy sources	K2	1	Lecture & Discussion	Blackboard & PPT
		Explain structure of electric power system	K2	1	Lecture	Blackboard & PPT
		Explain classification ofsubstations and switchgear devices	K2	2	Lecture	Blackboard & PPT
Number of hours Required				**6**		
UNIT-II: THERMAL POWER STATIONS						
2	Understand the principle of operation of different components of thermal power stations (K2)	Discuss the Schematic arrangement&Selection of site for thermal power station	K2	1	Lecture	Blackboard & PPT
		Discuss the general layout of a thermal power plant showingpaths of coal, steam, water, air, ash handling system	K2	2	Lecture	Blackboard & PPT
		Illustrate the components of the thermal power stations like Boilers, Superheaters, Economizers, electrostatic precipitators Condensers, feed water circuit, Coolingtowers and Chimney	K2	4	Lecture	Blackboard & PPT
Number of hours Required				**7**		
UNIT-III: NUCLEAR POWER STATIONS						
3	Understand the principle of operation of different components of nuclear power stations (K2)	Discuss the Location of nuclear power plant	K2	1	Lecture	Blackboard & PPT
		Explain the Working principle of nuclear fission, nuclear fuels, nuclear chain reaction	K2	2	Lecture	Blackboard & PPT
		Illustrate nuclear reactor Components: Moderators, Control rods, Reflectors and Coolants	K2	2	Lecture	Blackboard & PPT
		Discuss the types ofnuclear reactors	K2	1	Lecture	Blackboard & PPT
		Explain Radiation hazards and shielding, nuclear waste disposal	K2	1	Lecture	Blackboard & PPT
Number of hours Required				**7**		

UNIT – IV: HYDRO POWER STATIONS						
4	Understand the principle of operation of different components of hydro power stations (K2)	Discuss the Schematic arrangement, advantages and disadvantages	K2	2	Lecture	Blackboard & PPT
		Choice of site constituents of hydro power plant, Hydro turbine	K2	2	Lecture	Blackboard & PPT
		Discuss the Environmental aspects for selecting the sites and locations ofhydro power stations	K2	1	Lecture	Blackboard & PPT
Number of hours Required				**5**		
UNIT-V: SOLAR POWER PLANT						
5	Understand the working of solar photo voltaic systems and applications (K2)	Discuss the Solar photovoltaic cell, module, array	K2	1	Lecture	Blackboard& PPT
		Explain the construction of power plant&Efficiency of solar cells	K2	2	Lecture	Blackboard & PPT
		Discuss the CellI-V characteristics	K2	1	Lecture	Blackboard & PPT
		Explain Equivalent circuit of solar cell, Series resistance and Shunt resistance	K2	1	Lecture	Blackboard& PPT
		Discuss Applications and System design: storage sizing and PV system sizing	K2	2	Lecture	Blackboard& PPT
Number of hours Required				**7**		
UNIT-VI: WIND POWER PLANT						
6	Understand the wind energy conversion systems, efficiency and power generation (K2)	Explain Sources of wind energy - Wind patterns	K2	1	Lecture& Discussion	Blackboard & PPT
		Discuss the Types of turbines, Horizontal axis and vertical axismachines	K2	2	Lecture	Blackboard & PPT
		Explain the construction of power plant, Efficiency and Poweroutputofwindturbine	K2	2	Lecture	Blackboard & PPT
		Selectionofgenerator(synchronous,induction) and Explain Power generation for utility grids.	K2	2	Lecture	Blackboard & PPT
Number of hours Required				**7**		